

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B2: SYMMETRY AND RELATIVITY

TRINITY TERM 2022

Monday, 13th June, 2:30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Throughout this paper, the quotient rule of tensor analysis and the chain rule of partial derivatives may be quoted without proof.

1. Two frames are represented by Cartesian co-ordinate systems in standard configuration. The transformations of vector components between the frames are defined in the usual manner:

$$x'^{\mu} = \frac{\partial x'^{\mu}}{\partial x^{\nu}} x^{\nu}, \quad \text{with} \quad x'^0 = \gamma x^0 - \gamma \beta x^1, \quad x'^1 = \gamma x^1 - \gamma \beta x^0, \quad (1)$$

$$x'^2 = x^2, \quad x'^3 = x^3$$

and

$$x'_{\mu} = \frac{\partial x'_{\mu}}{\partial x_{\nu}} x_{\nu}, \quad \text{with} \quad x'_0 = \gamma x_0 + \gamma \beta x_1, \quad x'_1 = \gamma x_1 + \gamma \beta x_0, \quad (2)$$

$$x'_2 = x_2, \quad x'_3 = x_3,$$

where $x^{\mu} = (ct, x, y, z)$ and $x_{\mu} = (-ct, x, y, z)$.

(a) Consider the two operators

$$\left(\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \quad \text{and} \quad \left(-\frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right). \quad (3)$$

One of these is referred to as ∂^{μ} and the other ∂_{μ} . By deriving the transformation properties of the operators in (3) and comparing them to those of equations (1) and (2), state which is ∂_{μ} and which is ∂^{μ} . [You may use the chain rule and that $\frac{\partial x^a}{\partial x'^b}$, considered as a matrix, is the inverse of $\frac{\partial x'_a}{\partial x_b}$.] [5]

(b) Show that $\partial_{\mu} \partial^{\mu}$ is Lorentz invariant. [5]

(c) Show from Maxwell's equations that $\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{J} = 0$, where ρ is charge density and $\vec{J}$ is 3-current density. Given that this relation holds true in all frames of reference, find a 4-vector J^{μ} such that $\partial_{\mu} J^{\mu} = 0$. [3]

(d) Electric and magnetic fields can be computed from vector and scalar potentials using $\vec{E} = -\vec{\nabla} \phi - \frac{\partial \vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla} \times \vec{A}$. Let A^{μ} be a quantity whose components are $(\phi/c, \vec{A})$. Assuming the condition $\partial_{\mu} A^{\mu} = 0$, show from Maxwell's equations that $\partial^{\nu} \partial_{\nu} A^a = -\mu_0 J^a$, and prove that A^{μ} is a 4-vector. [8]

(e) An electromagnetic plane wave in otherwise empty space has the form $A^b = \epsilon^b \sin(k^{\mu} x_{\mu})$ where ϵ^b is a constant amplitude 4-vector and k^b is a constant 4-wave-vector. Show that in Lorenz gauge we can always choose ϵ^b such that $\vec{k} \cdot \vec{\epsilon} = 0$. [4]

2. Note: $c = 1$ throughout this problem.

(a) Consider a collection of N particles. Let $(E_i, \vec{p}_i)$ be the 4-momentum of the i 'th particle in the laboratory frame. Show that the velocity, relative to the laboratory, of the zero momentum frame (hereafter called the CM frame) for this collection of particles is given by

$$\vec{\beta} = \frac{-\sum_i^N \vec{p}_i}{\sum_i^N E_i}. \quad [5]$$

(b) A neutral pion (π^0) of 4-momentum $(E_\pi, \vec{p}_\pi)$ decays into two photons $(E_1, \vec{p}_1)$ and $(E_2, \vec{p}_2)$ in the lab frame. Let A be the angle between the directions of propagation of the photons in the lab frame. Show that

$$\sin^2\left(\frac{A}{2}\right) = \frac{M_\pi^2}{4E_1E_2},$$

where M_π is the mass of the pion. [5]

(c) For the decay process described in part (b), the emission is isotropic in the CM frame. This means the probability distribution is such that the differential number of photons in the direction θ_* satisfies $dN/d\theta_* = a \sin \theta_*$ where θ_* is the angle in the CM frame from some given direction, and a is a constant. Taking the given direction to be that of the boost from lab frame to CM frame, and defining A as before, show that

$$\frac{dN}{dA} = af(\gamma) \frac{\cos(A/2)}{\beta \sin^2(A/2)} \times \frac{1}{\sqrt{\gamma^2 \sin^2(A/2) - 1}}$$

where γ and β define the boost from the CM to the lab frame, and $f(\gamma)$ is a simple function of γ which your answer should specify. [10]

(d) What is the minimum value of the separation angle A , and at what value of A is the maximum observed in the distribution? [You do not need to differentiate the expression for $\frac{dN}{dA}$ for this part of the question.] [5]

3. (a) A moving particle is observed for a short interval dt of time in an inertial frame A. Show that the time elapsed in the particle's rest frame is $d\tau = dt/\gamma_u$, and that the components of the 4-velocity $u^\alpha = dx^\alpha/d\tau$ are $u^\alpha = (c\gamma_u, \gamma_u \vec{u})$ where $\vec{u}$ is the 3-velocity in frame A. [4]

(b) Find the components of the 3-velocity observed in a frame B in standard configuration with A (i.e. with aligned axes and moving at speed v in the positive x -direction relative to A). [5]

(c) Suppose now that frame B moves with velocity $\vec{v}$ in some arbitrary direction relative to frame A, but the axes remain aligned. Let $\vec{w}$ be the velocity of the particle in frame B. Show that

$$|\vec{w}| = \frac{\sqrt{(\vec{v} - \vec{u}) \cdot (\vec{v} - \vec{u}) - \frac{1}{c^2}(\vec{v} \times \vec{u}) \cdot (\vec{v} \times \vec{u})}}{1 - \vec{v} \cdot \vec{u}/c^2}.$$

[You may use the identity $(\vec{u} \times \vec{v}) \cdot (\vec{u} \times \vec{v}) = |\vec{u}|^2 |\vec{v}|^2 - (\vec{u} \cdot \vec{v})^2$] [8]

(d) Two stars S_1 and S_2 move at speed $c/2$ in frame A, one along the x direction and the other at 45 degrees to this, along the $(1, 1, 0)$ direction. It so happens that at some moment the light emitted by S_2 with frequency ω_0 in its rest frame is later received by S_1 with that same frequency as observed in the rest frame of S_1 . Find the angle between the direction of propagation of this light and the direction of the motion of S_2 , when both are observed by S_1 . [6]

Using a simple argument from symmetry, or otherwise, find the direction of propagation of this light in frame A. [2]

4. (a) A particle of rest mass m , initially at rest, is subject to a constant force $\vec{F}$. Find the velocity as a function of time. Hence show that the maximum attainable speed is bounded at $t = \infty$ and is equal to c . [You are given that force and velocity are related by

$$\vec{F} = \frac{d}{dt} (\gamma m \vec{v})$$

where $\gamma = (1 - v^2/c^2)^{-1/2}$.]

[3]

(b) Consider the four-force defined by $F^\alpha = dp^\alpha/d\tau$ and show that if $dm/dt = 0$ then this can be written as $F^\alpha = (\gamma \vec{F} \cdot \vec{v}/c, \gamma \vec{F})$ where

$$\vec{F} = m\gamma^3 \left(\vec{a} + \frac{\vec{v} \times (\vec{v} \times \vec{a})}{c^2} \right)$$

with $\vec{a} = d\vec{v}/dt$. [You may use $(\vec{A} \times \vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$]

[10]

(c) Power lost via radiation by an accelerating charge can be computed using Larmor's formula:

$$P = \frac{-1}{6\pi\epsilon_0} \frac{q^2}{m^2 c^3} \frac{dp^\alpha}{d\tau} \frac{dp_\alpha}{d\tau}.$$

[Note: you are not required to derive this formula, only use it.] Compare the radiative power loss for a charged particle in a circular accelerator to that for a particle of the same energy and acceleration in a linear accelerator, as a function of the Lorentz factor γ , giving the explicit dependence on γ .

[7]

(d) Assume you had a limited budget of energy to accelerate a charged particle but were free to use either a muon or an electron in either a linear or circular accelerator as long as you minimized the lost power. Which would you choose and why?

[5]